

GA1025 Macro Theory

Solution to Problem Set 3

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Problem 1.1

The sequence problem of the Cass-Koopmans model is given by

$$\begin{aligned} V^*(k_0) &= \max_{\{k_{t+1}\}_0^\infty} \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma} - 1}{1-\gamma} \\ \text{subject to } & k_{t+1} = (1-\delta)k_t + Ak_t^\alpha - c_t \\ & k_0 \text{ is given} \end{aligned}$$

After substituting the resource constraints, we have

$$V^*(k_0) = \max_{\{k_{t+1}\}_0^\infty} \sum_{t=0}^{\infty} \beta^t \frac{((1-\delta)k_t + Ak_t^\alpha - k_{t+1})^{1-\gamma} - 1}{1-\gamma}$$

Take the derivative w.r.t. k_{t+1} , we have

$$\beta^t \left[(1-\delta)k_t + Ak_t^\alpha - k_{t+1} \right]^{-\gamma} (-1) + \beta^{t+1} \left[(1-\delta)k_{t+1} + Ak_{t+1}^\alpha - k_{t+2} \right]^{-\gamma} \left(1 - \delta + A\alpha k_{t+1}^{\alpha-1} \right) = 0$$

After simplification, we have

$$\left[(1-\delta)k_t + Ak_t^\alpha - k_{t+1} \right]^{-\gamma} = \beta \left[(1-\delta)k_{t+1} + Ak_{t+1}^\alpha - k_{t+2} \right]^{-\gamma} \left(1 - \delta + A\alpha k_{t+1}^{\alpha-1} \right)$$

In the steady state, $k_t = k_{t+1} = k_{t+2} = \bar{k}$, as a result, we can compute the ss $\bar{k}$ as follows.

$$\bar{k} = \left(\frac{1 - \beta + \delta\beta}{A\alpha\beta} \right)^{\frac{1}{\alpha-1}}$$

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After substituting c_t back to the Euler equation, we have

$$c_t^{-\gamma} = \beta(1 - \delta + A\alpha k_{t+1}^{\alpha-1})c_{t+1}^{-\gamma}$$

The equation above implies the behavior of consumption smoothing, where the agent equalizes the marginal utility of spending today (LHS) and the marginal utility of spending tomorrow (RHS).

Problem 1.2

The FE formulation of the Cass-Koopmans model is given by

$$V(k) = \max_{k' \in \Gamma(k)} \frac{c^{1-\gamma} - 1}{1 - \gamma} + \beta V(k')$$

subject to $k' = (1 - \delta)k + Ak^\alpha - c$

Problem 1.3

I implemented VFI with the following computational parameters:

Computational parameters	Values
k_{min}	$0.95 \cdot k_{ss}$
k_{max}	$1.05 \cdot k_{ss}$
k_{vec}	$k_{min} : \frac{k_{max} - k_{min}}{I - 1} : k_{max}$
max_iter	1000 (maximum number of iterations)
tol	10^{-8} (tolerance for convergence)

Table 0.1: Summary of Parameters

The figures below display the policy function and value function.

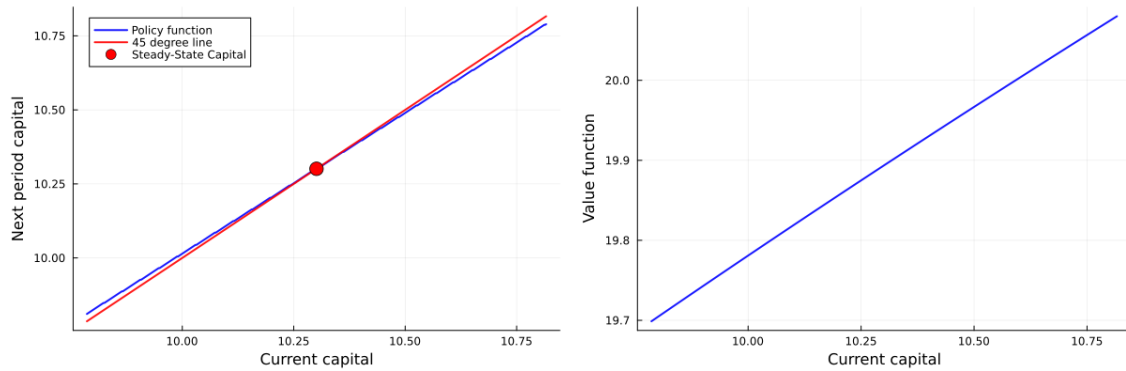


Figure 0.1: Value Function Iteration

Problem 1.4

After adjusting the grid, I obtain the same results as 1.3 (shown as follows).

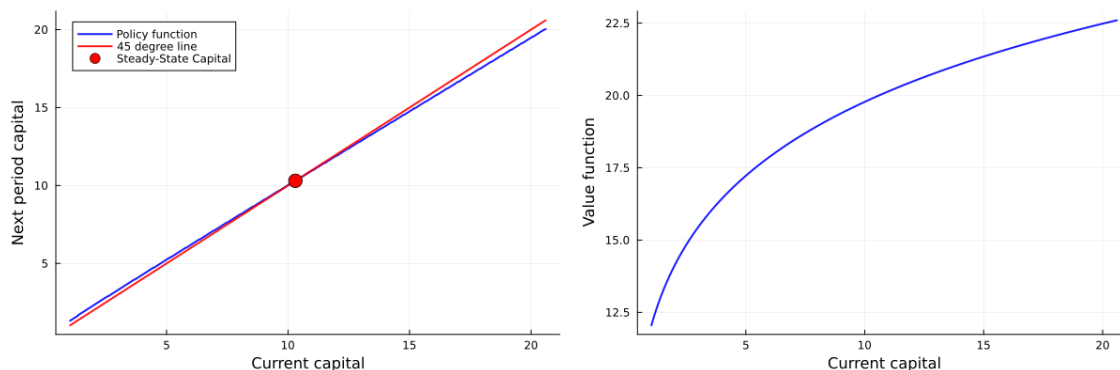


Figure 0.2: Value Function Iteration

Problem 1.5

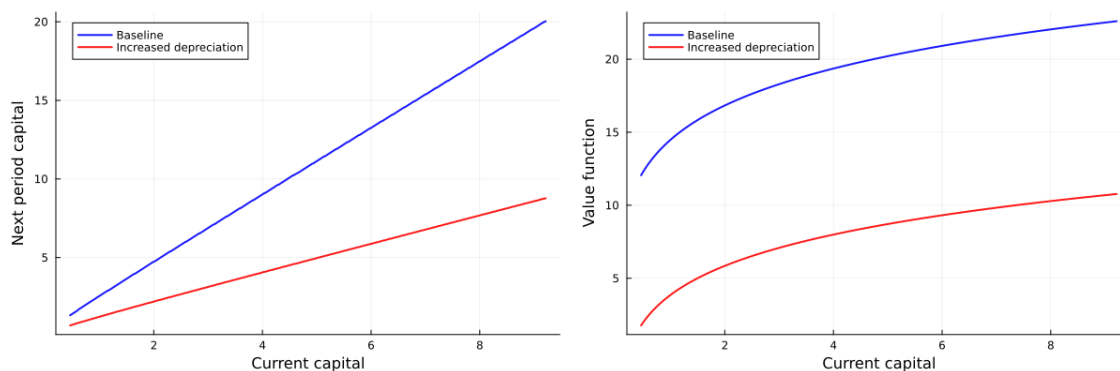


Figure 0.3: Increased capital share

Increased depreciation

More impatience Intuitively, the value function decreases for each k because the continuation values decrease given the flow utility. As the planner is less patient, she/he will consume more today and thus save less.

Remark: The speed of convergence of the VFI increases as β decreases because the value function approaches to the utility function, which is a deterministic object. In fact,

Increasing capital share Intuitively, the value function increases for each k because the marginal product of capital increases. For the same reason, higher MPK also incentivizes more savings.

Lower intertemporal elasticity of substitution

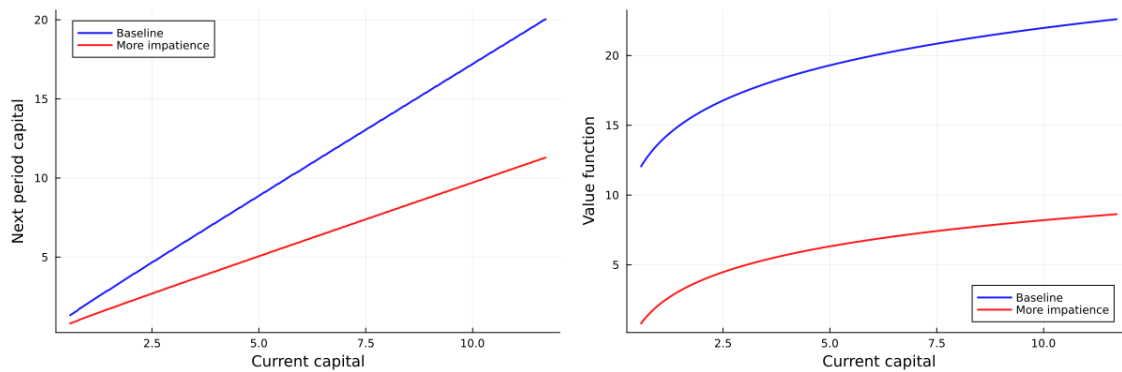


Figure 0.4: More Impatience

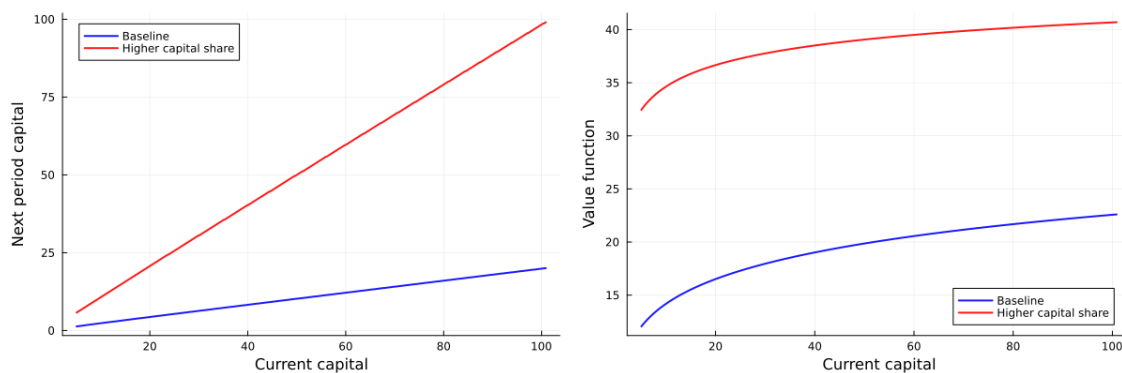


Figure 0.5: Increased capital share

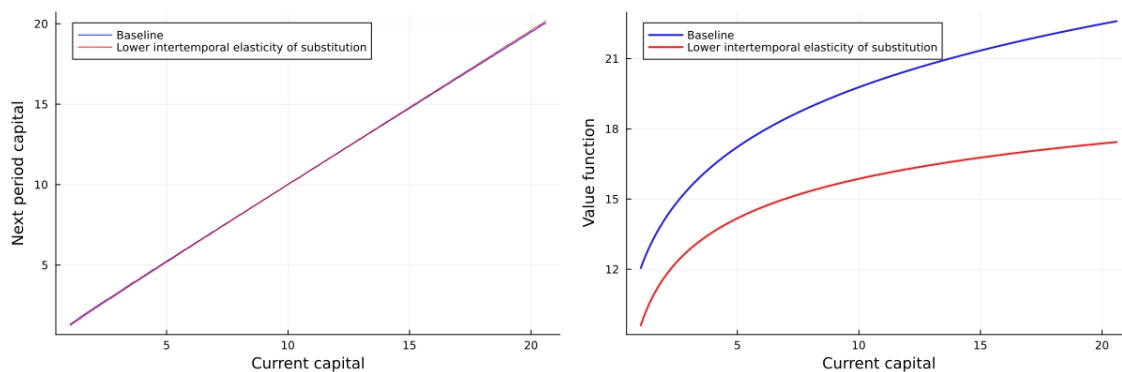


Figure 0.6: Increased capital share

Shooting Algorithm

Problem 1.6

The reward function is given by

$$F(x, y) = \frac{((1 - \delta)x + Ax^\alpha - y)^{1-\gamma} - 1}{1 - \gamma}$$

Therefore, the Bellman Equation is

$$V(k) = \max_{k' \in \Gamma(k)} F(k, k') + \beta V(k')$$

It follows that the Euler equation (using k_t, k_{t+1}, k_{t+2}) is

$$((1 - \delta)k_t + Ak_t^\alpha - k_{t+1})^{-\gamma} = \beta ((1 - \delta)k_{t+1} + Ak_{t+1}^\alpha - k_{t+2})^{-\gamma} (1 - \delta + A\alpha k_{t+1}^{\alpha-1})$$

and the transversality condition is

$$\lim_{t \rightarrow \infty} \beta^t ((1 - \delta)k_t + Ak_t^\alpha - k_{t+1})^{-\gamma} k_{t+1} = 0$$

Problem 1.7

Based on the Euler equation, we can express k_{t+2} as a function of k_t and k_{t+1} as follows.

$$k_{t+2} = (1 - \delta)k_{t+1} + Ak_{t+1}^\alpha - [\beta (1 - \delta + A\alpha k_{t+1}^{\alpha-1})]^{1/\gamma} ((1 - \delta)k_t + Ak_t^\alpha - k_{t+1})$$

Problem 1.8

Starting with $k_0 = 0.25k_{ss}$ and a search range between 2.8 and 3.0, I use the shooting algorithm to find the optimal k_1 . The trajectories of capital stocks and the transversality condition (in logarithm) are shown below.

Problem 1.9

The figure below compares the k_t trajectory derived from the Shooting algorithm to the VFI result. The figure shows that these paths align almost perfectly.

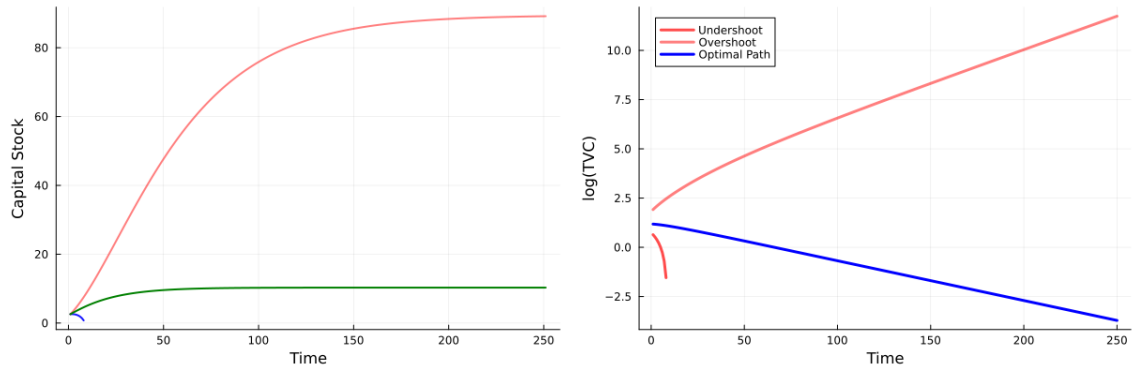


Figure 0.7: Trajectories of k_t and TVC

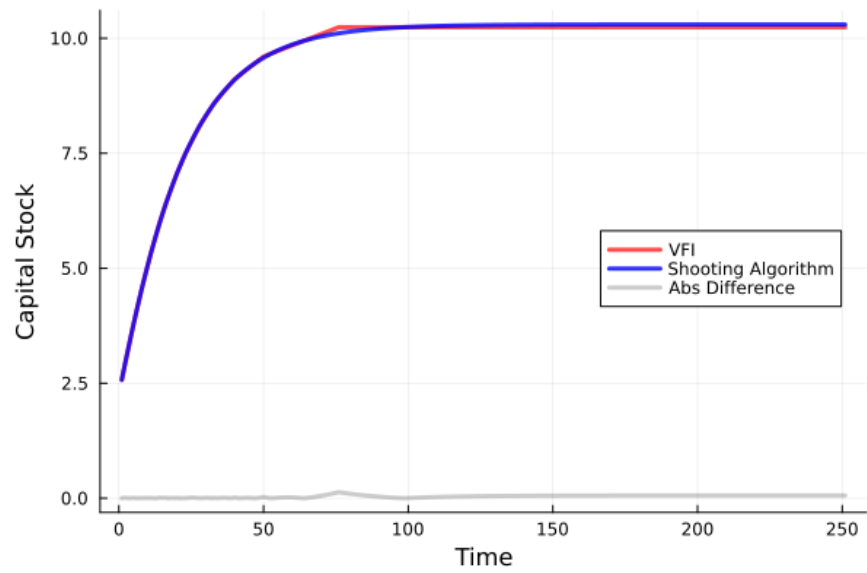


Figure 0.8: Comparison of k_t

Second Order Difference Equation

Problem 1.10

The objective function for the sequence problem in the adjustment model is given by

$$V^*(x_0) = \max_{\{x_{t+1}\}_0^\infty \in \mathbb{R}} \sum_{t=0}^{\infty} \beta^t \left[-\frac{\phi}{2} x_{t+1}^2 - \frac{\kappa}{2} (x_{t+1} - x_t)^2 \right]$$

Take the first-order condition with respect to x_{t+1} , we have

$$\beta^t [-\phi x_{t+1} - \kappa(x_{t+1} - x_t)] + \beta^{t+1} \kappa(x_{t+2} - x_{t+1}) = 0$$

After simplification, we have

$$\beta \kappa(x_{t+2} - x_{t+1}) = \phi x_{t+1} + \kappa(x_{t+1} - x_t) \quad (0.1)$$

According to the Definition, the transversality condition is given by

$$\lim_{t \rightarrow \infty} \beta^t \kappa(x_{t+1} - x_t) x_t = 0$$

Answer to Question 1.11

The characteristic function is given by

$$m^2 - \left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right) m + \frac{1}{\beta} = 0$$

It follows that the roots are

$$m_{1,2} = \frac{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right) \pm \sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}}}{2}$$

As a result, $m_1 \cdot m_2 = \frac{1}{\beta}$.

Answer to Question 1.12

Therefore, the general solution to the problem is given by

$$x_t = A \left(\frac{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right) - \sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}}}{2} \right)^t + B \left(\frac{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right) + \sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}}}{2} \right)^t$$

Let $m_1 < m_2$, since $m_1 \cdot m_2 = \frac{1}{\beta}$, we have

$$\begin{aligned} m_1 &< \frac{1}{\beta m_1} \\ \Rightarrow m_1 &< \beta^{-1/2} \quad \text{and} \quad m_2 > \beta^{-1/2} \end{aligned}$$

Then, we can express m_1 and m_2 as

$$\begin{aligned} m_1 &= \beta^{-1/2} \epsilon, \quad \epsilon \in (0, 1) \\ m_2 &= \beta^{-1/2} \delta, \quad \delta > 1 \end{aligned}$$

Substituting x_t with $A(\beta^{-1/2}\epsilon)^t + B(\beta^{-1/2}\delta)^t$, the Transversality condition becomes

$$\begin{aligned} &\lim_{t \rightarrow \infty} \beta^t \left[A(\beta^{-1/2}\epsilon)^{t+1} + B(\beta^{-1/2}\delta)^{t+1} - A(\beta^{-1/2}\epsilon)^t - B(\beta^{-1/2}\delta)^t \right] \cdot \left[A(\beta^{-1/2}\epsilon)^t + B(\beta^{-1/2}\delta)^t \right] = \\ &\lim_{t \rightarrow \infty} \beta^t \left[A(\beta^{-1/2}\epsilon)^t (\beta^{-1/2}\epsilon - 1) + B(\beta^{-1/2}\delta)^t (\beta^{-1/2}\delta - 1) \right] \cdot \left[A(\beta^{-1/2}\epsilon)^t + B(\beta^{-1/2}\delta)^t \right] \leq \\ &\lim_{t \rightarrow \infty} A^2 \epsilon^{2t} (\beta^{-1/2}\epsilon - 1) + B^2 \delta^{2t} (\beta^{-1/2}\delta - 1) \end{aligned}$$

Because $\delta > 1$, $\delta^{2t} \rightarrow \infty$, therefore, this equation converges to zero if $B = 0$, where B is the coefficient of the larger root.

Answer to Question 1.13

Since the initial condition is $x_0 = Am^0 = A$, we have $A = x_0$.

Answer to Question 1.14

From 1.13, we have

$$x_t = x_0 \left(\frac{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right) - \sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}}}{2} \right)^t$$

Take the first-order condition with respect to κ for ϕ ,

$$\frac{\partial x_t}{\partial \kappa} = \frac{x_0}{2^t} t \left(\frac{\phi + \kappa}{\beta \kappa} + 1 - \sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}} \right)^{t-1} \frac{\phi}{\beta \kappa^2} \left(\frac{1}{\sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}}} - 1 \right)$$

and

$$\frac{\partial x_t}{\partial \phi} = \frac{x_0}{2^t} t \left(\frac{\phi + \kappa}{\beta \kappa} + 1 - \sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}} \right)^{t-1} \frac{1}{\beta \kappa} \left(1 - \frac{1}{\sqrt{\left(\frac{\phi + \kappa}{\beta \kappa} + 1 \right)^2 - \frac{4}{\beta}}} \right)$$

Intuitively, as κ increases, the adjustment cost increases, and thus the agent adjusts x_t more slowly. Conversely, as ϕ increases, the marginal cost of holding x_t increases, and thus the agent adjusts x_t more quickly.

Guess and Verify

Answer to Question 1.15

Incorporating the budget constraint, we can rewrite the model as follows.

$$\max_{k_{t+1} \in \Gamma(k_t)} \sum_{t=0}^{\infty} \beta^t U(G(k_t) + (1 - \delta)k_t - k_{t+1})$$

where $\Gamma(k_t) = [0, G(k_t) + (1 - \delta)k_t]$

Therefore, the Bellman equation is given by

$$V(k) = \max_{k'} U(G(k) + (1 - \delta)k - k') + \beta V(k')$$

Answer to Question 1.16

If $U(c) = \log(c)$, $G(k) = k^\alpha$, and $\delta = 1$, and we “guess” the value function is $V(k) = a + b \log k$, the Bellman equation formulated above becomes

$$a + b \log k = \max_{k'} \log(k^\alpha - k') + \beta(a + b \log k')$$

Then,

$$[k'] = -\frac{1}{k^\alpha - k'} + \beta \frac{b}{k'} = 0$$

As a result, the policy function is

$$k' = \frac{\beta b k^\alpha}{1 + \beta b}$$

Substituting $k' = \frac{\beta b k^\alpha}{1 + \beta b}$ back to the Bellman equation, we have

$$a + b \log k = \log \left(k^\alpha - \frac{\beta b k^\alpha}{1 + \beta b} \right) + \beta a + \beta \log \left(\frac{\beta b k^\alpha}{1 + \beta b} \right)$$

By comparing coefficients, we have

$$b = \frac{\alpha}{1 - \alpha\beta}$$

$$a = \frac{1}{1 - \beta} \left[\frac{1}{1 - \alpha\beta} \log(1 - \alpha\beta) + \frac{\alpha\beta}{1 - \alpha\beta} \log \frac{\alpha\beta}{1 - \alpha\beta} \right]$$

Answer to Question 1.17

I verify the analytical solutions above by comparing them to the numerical solutions obtained from the VFI. The figures below display the comparisons.

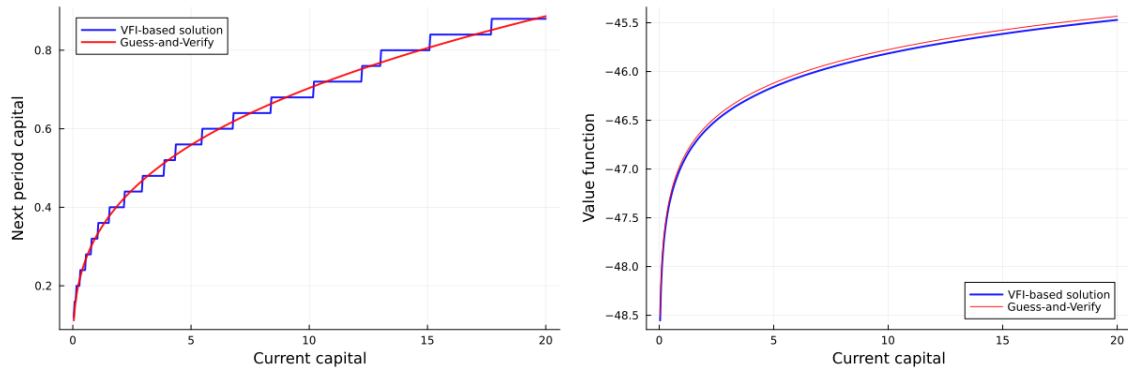


Figure 0.9: Guess-and-Verify

2. Probability Theory

Exercise 2.1

In a stochastic environment of a fair dice, the filtered probability space is given by

- $\Omega := \{1, 2, 3, 4, 5, 6\}^\infty$ denotes the sample space.
- $\mathcal{F} := 2^\Omega$
- $\{\mathcal{F}_t\}_t^\infty$ is defined as follows:

$$\mathcal{F}_1 = 2^{\{1,2,3,4,5,6\}}$$

...

$$\mathcal{F}_t = 2^{\{1,2,3,4,5,6\}^t}$$

...

$$\mathcal{F}_\infty = 2^{\{1,2,3,4,5,6\}^\infty}$$

with $\mathcal{F}_t \subseteq \mathcal{F}_{t+s}$ for any $s \geq 0$. Economically, $\mathcal{F}_t$ contains the information up to time t .

- $P = \frac{1}{2^\infty}$

Exercise 2.2

Q1. The sample space is given by $\Omega = \{H, T\}^T$. The sample space contains 2^T elements.

Q2. The sigma-algebra is given by $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}^T$.

Q3. Take $T = 4$, and $s^2 = \{H, T\}$, $F(s^t)$ includes the following

1. (H, T, H, H)
2. (H, T, H, T)
3. (H, T, T, T)
4. (H, T, T, H)

Q4. Note that for a fixed partial path s^t , there are 2^{T-t} possibilities. Therefore, the probability measure P representing fair iid coin flips is given by

$$P(s^T) = \frac{1}{2^T}$$

$$P(F(s^t)) = \frac{2^{T-t}}{2^T} = \frac{1}{2^t}$$

Q5. For $T = 3$,

$$\mathcal{F}_0 = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

$$\begin{aligned} \mathcal{F}_1 = & \{\emptyset, \\ & \{(H, H)\}, \{(H, T)\}, \{(T, H)\}, \{(T, T)\}, \\ & \{(H, H), (H, T)\}, \\ & \{(H, H), (T, H)\}, \\ & \{(H, H), (T, T)\}, \\ & \{(H, T), (T, H)\}, \\ & \{(H, T), (T, T)\}, \\ & \{(T, H), (T, T)\}, \\ & \{(H, H), (H, T), (T, H)\}, \\ & \{(H, H), (H, T), (T, T)\}, \\ & \{(H, H), (T, H), (T, T)\}, \\ & \{(H, T), (T, H), (T, T)\}, \\ & \{(H, H), (H, T), (T, H), (T, T)\}\} \end{aligned}$$

Q6. Consider two paths that share the same partial histories up to time t . Intuitively, their information overlap with each other. Therefore, we cannot assign probability to only one of them, instead, we need to partition the s^∞ based on their partial histories F_t , and assign probabilities according to F_t .

Q7.

1. The smallest σ -algebra $\mathcal{F}$ is $\{\emptyset, \Omega, \{s_0, s_1, H, s_3, \dots\}, \{s_0, s_1, T, s_3, \dots\}\}$
2. Denote $G_t := \{s_0, s_1, \dots : \sum_{i=1}^{10} s_i 1(s_i = H) = t\}$. Then

$$\mathcal{F} = \bigcup_{t=1}^{10} \{\emptyset, \Omega, 2^{G_t}\}$$

3. Denote $G'_t := \{s_0, s_1, \dots : \sum_{i < t} s_i 1(s_i = H) = 0\}$. Then

$$\mathcal{F} = \bigcup_t \{\emptyset, \Omega, 2^{G'_t}\}$$

Remark: The answers above does not depend on whether T is finite or infinite as long as

$T > 10$ for the second question.

Q8.

1. $\mathbb{E}[x] = 1/2$. $\mathbb{E}[x|\mathcal{F}_1] = 1/2$. $\mathbb{E}[x|\mathcal{F}_2] = x$.

2. The unconditional and conditional expectations are given by

$$\begin{aligned}\mathbb{E}[x] &= \sum_{t=1}^{10} t \frac{\binom{10}{t}}{2^{10}} \\ \mathbb{E}[x|s_1 = H] &= \sum_{t=1}^{10} t \frac{\binom{9}{t-1}}{2^9} \\ \mathbb{E}[x|s_1 = T] &= \sum_{t=1}^9 t \frac{\binom{9}{t-1}}{2^9} \\ \mathbb{E}[x|s_1, s_2 = H, H] &= \sum_{t=2}^{10} t \frac{\binom{8}{t-2}}{2^8} \\ \mathbb{E}[x|s_1, s_2 = H, T] &= \sum_{t=1}^9 t \frac{\binom{8}{t-1}}{2^8} \\ \mathbb{E}[x|s_1, s_2 = T, H] &= \sum_{t=1}^9 t \frac{\binom{8}{t-1}}{2^8} \\ \mathbb{E}[x|s_1, s_2 = T, T] &= \sum_{t=1}^8 t \frac{\binom{8}{t-1}}{2^8}\end{aligned}$$

3. The unconditional and conditional expectations are given by

$$\begin{aligned}\mathbb{E}[x] &= \sum_{t=1}^{\infty} t \left(\frac{1}{2}\right)^t = 2 \\ \mathbb{E}[x|s_1 = H] &= 0 \\ \mathbb{E}[x|s_1 = T] &= \sum_{t=2}^{\infty} t \left(\frac{1}{2}\right)^{t-1} = 3 \\ \mathbb{E}[x|s_1, s_2 = H, H] &= 0 \\ \mathbb{E}[x|s_1, s_2 = H, T] &= 0 \\ \mathbb{E}[x|s_1, s_2 = T, H] &= 0 \\ \mathbb{E}[x|s_1, s_2 = T, T] &= 4\end{aligned}$$

Q9. The Radon-Nikodym derivative of $\tilde{P}$ with respect to P for any partial history s^t is given by

$$\begin{aligned} h(s^t) &= \frac{\sum_{s^{t+j}|s^t} \tilde{P}(s^{t+j})}{P(s^t)} \\ &= \frac{\prod_{i=1}^t p^{(\# \text{heads in } s^i)} (1-p)^{(\# \text{tails in } s^i)}}{(1/2)^t} \end{aligned}$$

Q10. Under the fair specification, $(\mathcal{F}, P)$, w.l.o.g, consider $s = t + 1$

$$\mathbb{E}[x_{t+1}|\mathcal{F}_t] = x_t + p + (1-p)(-1)$$

which is equal to x_t iff $p = 1 - p$. Therefore, x_t is a martingale with respect to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ and P , but not with respect to $\{\mathcal{F}_t\}_{t \geq 0}$ and $\tilde{P}$,

Q11. The stochastic process in Q10 is time-invariant because

$$\mathbb{E}[x_{t+1}|x] = x + p + (1-p)(-1)$$

which only depends on x not t .

Q12. If $\hat{P}(s_t = H|s_{t-1} = H) = \hat{P}(s_t = T|s_{t-1} = T) = p$, then

$$\begin{aligned} \mathbb{E}[x_{t+1}|x_t] &= x_t + p + (1-p)(-1) \text{ if } s_t = H \\ \mathbb{E}[x_{t+1}|x_t] &= x_t + (1-p) + p(-1) \text{ if } s_t = T \end{aligned}$$

Therefore, $\mathbb{E}[x_{t+1}|x_t] = x_t$ if $p = 1/2$.